

EN 607: Modeling and Simulation of Energy Systems

Course Code	EN 607
Course Name	Modeling and Simulation of Energy Systems
L-T-P-C	2-0-2-3
Prerequisites	IC 110 (Engineering Mathematics)/ Instructor Consent
Intended for	B.Tech/M.Tech/MS/Ph.D.
Distribution	compulsory for M. Tech in Energy Engineering and elective for others
Semester	Odd/Even
Approval	9th Senate (May 2016), in the M.Tech Energy Engineering (Mechanical Design) curriculum

Preamble

The course is aimed to provide the students with general tools of modelling and simulation as applied to energy systems.

Course Modules

- **Physical Modelling:** Dimensional analysis and dimensionless groups, Similarity criteria, development of physical models based on similarity and dimensionless group. Application of physical modelling to energy systems (6 contact hours)
- **Mathematical modelling:** Continuous time dynamic modelling through differential equation, lumped parameter and distributed parameter models, Input-output models, transfer function, state space models and their transfer functions, models for control, transient and steady state response, stability; Simulation of models by MATLAB, applications of dynamic modelling and simulation to energy management in ships and PV systems. (8 contact hours)
- **Fuzzy logic Modelling:** Fuzzy sets, membership function, operations with fuzzy sets, Fuzzy knowledge, Fuzzy inference systems, fuzzy approach to identify power transformer criticality, data clustering to evolve fuzzy models, design of fuzzy logic controller for optimal control of boiler-turbine system. (5 contact hours)
- **Artificial neural network (ANN) modelling:** Neuron, Neural network and its architecture, training, validation and test of the back propagation network (BPN), modelling of solar distillation system, ANFIS models. (6 contact hours)
- **Non-traditional techniques of optimization:** simulated annealing and genetic algorithm. (3 contact hours)

Text Books

1. *Similarity and Modelling in Science and Engineering* by Josef Kunes, Cambridge International Science Publishing, 2012.
2. *Modern Control Engineering* by Katsuhiko Ogata, Prentice Hall India,
3. *Fuzzy logic with Engineering Applications* by Timothy J. Ross, John Wiley & Sons, 2010
4. *Neural Networks* by Simon Haykin, Pearson Education, Inc, 1999
5. *Optimization for Engineering Design* by K. Deb, Prentice Hall India Pvt. Ltd., 1991